

Research Proposal: Online Feedback Distillation: Teaching Lightweight to Mimic Expert Feedback in Reasoning Tasks

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Abstract

We propose an adaptive distillation framework in which a lightweight student model learns expert-level feedback by leveraging larger models. The expert model not only generates expert-level feedback but also produces answers and refines them based on unified feedback from both amateur and expert feedbacks. Ultimately, our goal is to reduce reliance on costly expert queries while improving efficiency, scalability, and feedback quality in reasoning systems.

1 Introduction

1.1 Motivation

Many current reasoning systems rely on expert models such as GPT-4 to generate feedback, but repeatedly querying these models is expensive and inefficient Madaan et al. [2023]. Frameworks like CLEAR (Contrasting Textual Feedback with Experts and Amateurs for Reasoning) attempt to reduce this cost by incorporating amateur models to contrast expert feedback Rufail et al. [2025]. However, these amateur models often provide low-quality, noisy feedback and cannot be relied upon for consistent improvements. As a result, CLEAR systems remain heavily dependent on expert models, which introduces bias towards expert feedback and limits the contribution of amateur models. Additionally, the base model that generates answers lacks the ability to self-critique and instead relies solely on external feedback from amateur and expert sources. This limitation is evident in methods like CLEAR, Chain-of-Draft, or Chain-of-Thought when the model depends exclusively on external feedback rather than self-critiquing its own text generation.

We propose a feedback distillation framework in which a large model serves as both the base model (which solves the task) and the teacher model, generating expert-level feedback. Our approach involves only two models: the expert model and a single adaptive student model. The teacher/expert model combines the roles of the base and expert models by both performing the task and generating expert feedback, alongside providing the guidance to the student model to improve its performance through iterative feedback distillation.

This shift from fixed amateurs to adaptive students has several advantages, which improves convergence in iterative refinement tasks by stabilizing the feedback signal, enhances generalization in settings with limited expert data, and mitigates the risk of amplifying biases

present in the expert model by training on a more diverse set of feedback examples. According to Yao et al. [2023], student models distilled from expert reasoning can achieve near-expert performance with a far lower computational cost, which makes them suited for deployment in resource-constrained environments. Furthermore, Madaan et al. [2023] show that iterative feedback loops—where a model critiques and refines its own output—result in continual improvements in both reasoning accuracy and overall task performance.

We believe that our approach is well-founded. KD (Knowledge Distillation) is a widely used technique that enables small models to approximate the behavior of larger ones. By integrating it into a feedback loop and combining it with task-time updates, we allow the student model to continuously refine its guidance without costly expert reliance. Previous work like CLEAR and BeCLEAR (best-first search algorithm with CLEAR) already demonstrate the benefits of structured feedback, and we aim to build on that by replacing static amateurs with dynamic, distilled students.

Furthermore, evidence from adjacent domains supports our approach. For example, the paper *Smaller but Better: Self-Paced Knowledge Distillation for Lightweight yet Effective LCMs* [Chen et al., 2024] applies a self-paced knowledge distillation framework to train compact models for code generation, improving student model performance by prioritizing high-quality training signals. Their student models, trained through iterative feedback and scoring mechanisms, outperformed much larger models like ChatGPT on HumanEval. While their focus is not reasoning, the success of real-time, feedback-driven student training lends support to our hypothesis that lightweight models can deliver expert-quality feedback through targeted, efficient updates.

1.2 Contribution

Our work introduces the first application of knowledge distillation to feedback-driven reasoning systems like CLEAR and BeCLEAR. While these prior frameworks rely on contrasting fixed amateur and expert feedback during inference, we adopt a more dynamic approach: we actively train the student model in real time to imitate expert feedback. Additionally, the expert model serves as a base model that not only performs the task but also self-critiques and revises its answer to produce improved final outputs.

This offers several key contributions:

- We integrate online knowledge distillation into feedback refinement loops, the first in the context of CLEAR.
- We treat the teacher model not only as an active feedback proposer but also the base model responsible for generating the final output, rather than just a contrasting agent.
- We reduce reliance on expert queries while maintaining or improving performance.
- Our student evolves during task execution, improving robustness and generalization.

This adaptive framework allows our system to scale to real-world, resource-constrained environments, something traditional expert-amateur systems struggle with. It also enables stronger convergence during iterative refinement and smoother optimization, especially on tasks that require precise reasoning like GSM8K Cobbe et al. [2021a](math word problems) and CommonGenHard Lin et al. [2020](constrained generation).

The research question we aim to answer is: Can we improve feedback quality in expert-amateur feedback frameworks by enabling the amateur model to learn from expert feedback online during task execution—thereby producing better revised answers than the initial ones – instead of relying on fixed, noisy amateur models and repeatedly querying the expert?

We evaluate our method by comparing task accuracy, convergence speed, and efficiency against:

- Traditional fine-tuning and prompting techniques, including Chain-of-Thought (CoT) and Chain-of-Doubt (CoD).
- The predecessor feedback-based reasoning framework CLEAR.

1.3 Related Literature

- Rufail et al. introduce CLEAR and BeCLEAR, two advanced reasoning techniques that outperform existing methods like Chain of Thought (CoT) and Tree of Thoughts (ToT) while also effectively reducing toxicity in generated outputs.
- Chen et al. introduce SODA (Self-Paced knowledge Distillation), a curriculum-based framework designed to train lightweight yet effective code generation models. SODA enhances student model performance by combining contrastive learning

with adaptive feedback loops that evolve training difficulty based on model performance. The approach achieves superior or comparable results to much larger models like ChatGPT on benchmarks such as HumanEval.

2 Experimental Setup

Datasets

We select benchmark datasets covering a range of tasks requiring feedback-driven or iterative reasoning.

- **Math Reasoning:** GSM8K Cobbe et al. [2021b] — a dataset of multistep arithmetic and word problems widely used for evaluating reasoning and knowledge distillation.
- **Instruction Following:** Alpaca CRFM [2023] — a dataset for instruction-following tasks, commonly used to evaluate model adaptability and instruction comprehension.
- **Reading Comprehension:** DROP Dua et al. [2019] — a dataset requiring discrete reasoning over paragraphs, including arithmetic and complex reading comprehension questions.

CLEAR Baseline Evaluation

To benchmark our feedback distillation framework, we implement and evaluate the CLEAR baseline Rufail et al. [2025] using an open-weight LLaMA-3-8B-Instruct model as both the expert feedback model and the base model. In this setup, there are three models: the base model, the amateur model, and the expert model. The base model is responsible for generating an initial answer to an input prompt. Next, both the amateur and expert models provide feedback on the base model’s answer, and the base model revises its response based on this feedback. This process replicates the standard CLEAR pipeline, where the base model’s initial output is iteratively refined through structured feedback from both expert and amateur providers.

Experimental Setup. We conduct experiments using the CLEAR system on the datasets described above. The procedure is as follows:

- The base model generates an initial answer.
- The amateur and expert models independently generate feedback on this answer.
- Feedback is merged via a contrastive summarization prompt that prioritizes expert insights while incorporating constructive contributions from the amateur model.
- The base model revises its answer based on the merged feedback and then performs a self-critique to guide a subsequent refinement.

Chain-of-Draft (CoD) Baseline Evaluation

To assess the effectiveness of iterative refinement via structured prompting, we implement a Chain-of-Draft (CoD) baseline using the LLaMA-3-8B-Instruct model. In this setup, we employ an engineered prompt to guide the model in constructing an outline during its initial assessment of the answer, with detailed evaluation criteria explicitly listed in the prompt. The model then analyzes the answer according to these criteria and provides a final assessment in the form of a one-paragraph feedback summary.

Experimental Setup The Chain-of-Draft (CoD) pipeline proceeds as follows:

- The base model generates an initial answer.
- The CoD model generates detailed feedback based on the initial answer.
- The base model revises its answer by incorporating the feedback.

Chain-of-Thought (CoT) Baseline Evaluation

To evaluate the effectiveness of iterative reasoning and feedback-driven self-improvement, we implement a Chain-of-Thought (CoT) baseline using the LLaMA-3-8B-Instruct model. In this setup, the CoT model is encouraged to articulate intermediate reasoning steps before providing feedback on the answer generated by the base model.

Experimental Setup We conduct experiments using the CoT system on the datasets described above. The procedure is as follows:

- The base model generates an initial answer.
- The CoT model then generates feedback on the base model’s answer, using prompt engineering to encourage step-by-step reasoning.
- The base model revises its answer by incorporating the feedback.

Evaluation Metrics To assess the quality of CoT refinement and feedback generation, we compute:

- **Answer Quality:** BERTScore, BLEU, ROUGE-L, and exact match accuracy (for classification and reasoning tasks).
- **Feedback Similarity:** Cosine similarity between expert and amateur feedback using MiniLM sentence embeddings.
- **Score Fidelity:** Mean squared error between student- and expert-assigned quality scores.
- **Toxicity:** Toxicity levels are computed for expert feedback, amateur feedback, and final student outputs using Detoxify.

- **Pareto Efficiency:** A 4D loss vector (scoring, semantic, LM loss, hidden-state similarity) is used to determine if student feedback lies on the Pareto frontier.

Task Setup

We select benchmark tasks that require iterative refinement. For each task, the process is as follows:

1. Prepare an initial input prompt I .
2. Generate an initial output x_0 using the expert model.
3. Use the expert model to generate the expert feedback prompt p_{expert} based on x_0 and I .
4. Use the amateur model to generate the amateur feedback prompt p_{amateur} based on x_0 and I .
5. Use the expert model to generate unified feedback by combining p_{expert} and p_{amateur} .
6. The expert model applies the unified feedback to generate a revised answer.
7. The expert model then self-critiques the revised answer and produces the final output.

Iterative Feedback Scoring

Iterative Feedback Scoring relies on a learned feedback scoring mechanism that evaluates the quality of feedback generated by a student model by comparing it against expert feedback. During training, expert feedback is available, allowing the scoring model to measure alignment, often via cosine similarity, between student and expert feedback. This scoring serves as a learned loss function that guides the student model’s improvement. By minimizing this loss, the student model refines its feedback generation, enabling effective adaptation and better generalization even when ground truth labels are unavailable on new tasks.

Knowledge Distillation Update

We update the student model online to minimize divergence between p_{student} and p_{expert} , using:

- Embedding-based similarity (e.g., cosine similarity).
- Intermediate representation alignment.
- Real-time parameter updates during task execution.

Refined Output Generation

The refined output x_1 is generated by the expert model, which functions as both the feedback generator and the base model. It first produces its own feedback prompt p_{teacher} , then uses the combined feedback—comprising both p_{student} and the expert feedback p_{expert} —together with the input I to generate x_1 . Finally, the expert model self-critiques and produces the final answer.

Iterative Refinement Loop

This process is repeated iteratively: at each step, the student updates its feedback and output based on the evolving input-feedback context. The loop continues until a predefined stopping condition, determined using the Pareto frontier, is met. This approach enables dynamic adaptation and progressively refined reasoning over time.

Evaluation

We evaluate using the following metrics:

- **Task Accuracy:** Final output correctness.
- **Convergence Speed:** Number of refinement steps to reach stable output.
- **Computational Cost:** Resources required per iteration.
- **Feedback Quality:** Measured via similarity to expert feedback.
 - **ROUGE-L:** Captures overlapping longest common subsequences to evaluate fluency and coherence.
 - **Cosine Similarity:** Measures alignment between expert and student feedback using MiniLM embeddings.
 - **Toxicity Scores:** Computed using Detoxify for each feedback and refined response.

We compare our method against established baselines, including CLEAR and BeCLEAR introduced by Rufail et al., CoT by Wei et al. Wei et al. [2022]

- Despite mitigation efforts, the student model may slightly amplify biases present in expert models, as it learns directly from expert feedback.
- Online distillation and iterative refinement require significant computational resources, particularly with larger models and many iterations.
- If the expert feedback is biased or low-quality, the student model may inherit these issues.
- Feedback scoring metrics such as cosine similarity may not fully capture the nuance of natural language guidance.
- Future work should explore human-in-the-loop filtering and adversarial robustness testing.

3 Expected Results

- Our method successfully trains a student model that not only proposes prompts approximating expert feedback but also serves as the base model for final output generation.
- The unified student model significantly reduces computational cost and inference latency by eliminating repeated expert queries and fixed amateur contrasts.
- Student-generated feedback facilitates smoother and more efficient iterative refinement.
- This results in improved task accuracy with fewer refinement steps.
- Compared to baselines, our approach achieves comparable or superior final performance while being more resource-efficient.

4 Limitations

- Under training or insufficient feedback quality can limit the student’s ability to approximate expert feedback.

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